

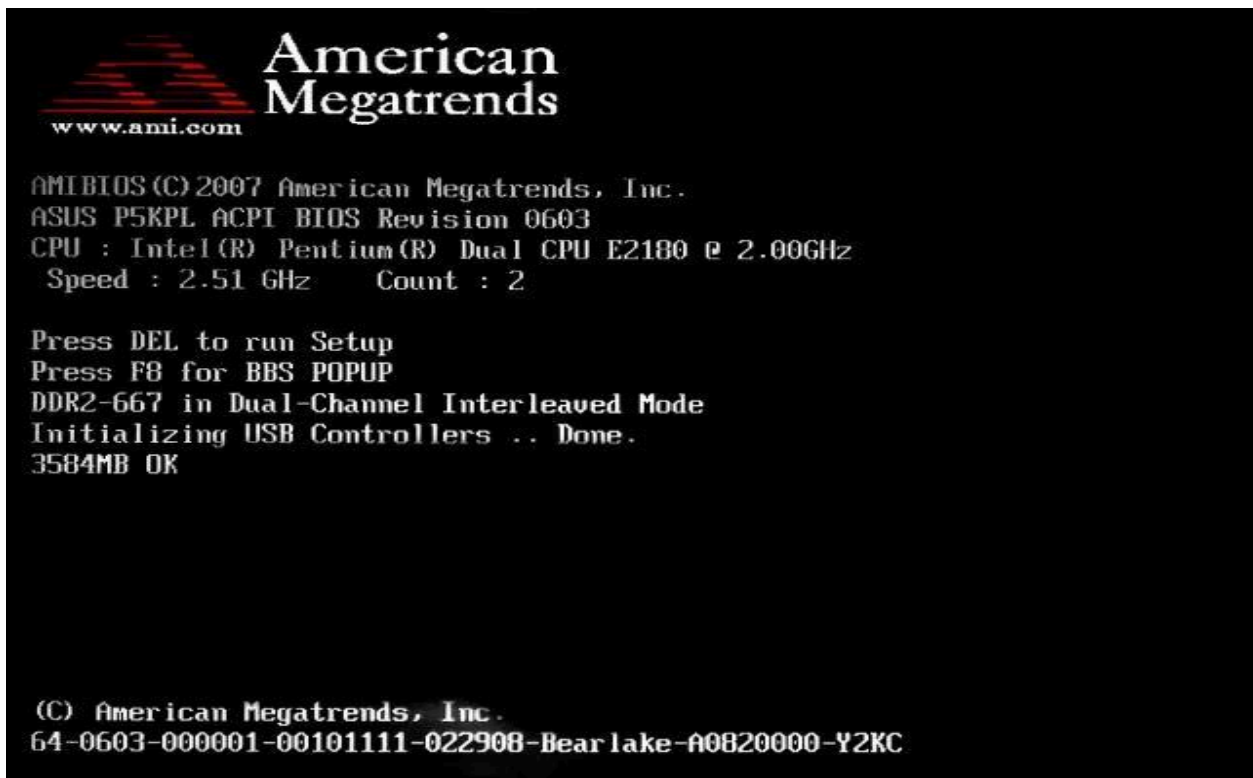
Linux Boot Process

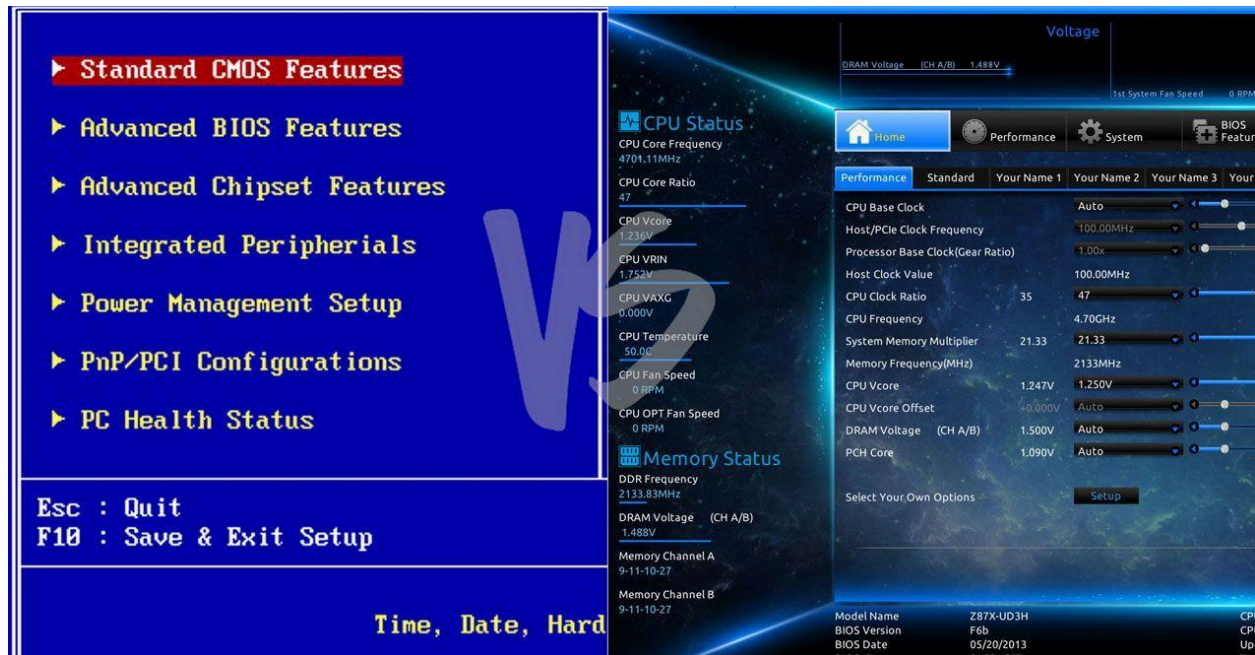
The **Linux boot process** is the sequence of steps that occur from the moment a computer is powered on until the Linux operating system is fully loaded and ready for users.

The process mainly consists of **6 stages**:

1. BIOS / UEFI
 2. Bootloader (GRUB)
 3. Kernel Initialization
 4. init / systemd
 5. System Initialization
 6. Login Prompt
-

1. BIOS / UEFI Stage





PhoenixBIOS 4.0 Release 6.0
 Copyright 1985-1999 Phoenix Technologies Ltd.
 All Rights Reserved
 Copyright 1996-1999 Intel Corporation.
 4S4EB2X0.05A.0009.P08

Micron Electronics, Inc.

Intel(R) Pentium(R) III processor 450 Mhz
 640K System RAM Passed
 255M Extended RAM Passed
 512K Cache SRAM Passed

Press F2 to Enter Setup

BIOS (Basic Input Output System)

- BIOS is firmware stored on the **motherboard chip**.
- It runs immediately when the computer is powered on.

Responsibilities

1. Performs **POST (Power-On Self Test)**
 - Checks hardware components like:
 - RAM
 - CPU
 - Keyboard
 - Disk drives
2. Detects **bootable devices**
 - Hard disk
 - SSD
 - USB
 - CD/DVD
3. Searches for **MBR (Master Boot Record)** in the first sector of the boot disk.

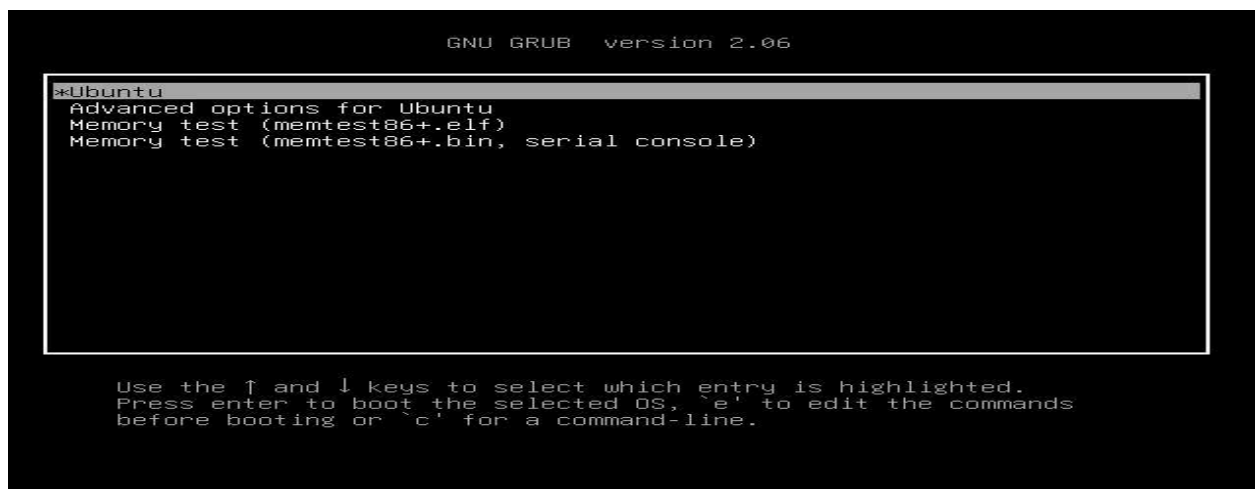
Boot Order Example

1. CD/DVD
2. USB
3. Hard Disk
4. Network

UEFI (Modern Replacement for BIOS)

- Supports **larger disks**
 - Faster boot
 - Uses **GPT partitioning**
 - Secure boot support
-

2. Bootloader Stage (GRUB)



GNU GRUB version 1.99,5.11.0.175.1.0.0.13.18988

Minimal BASH-like line editing is supported. For the first word, TAB lists possible command completions. Anywhere else TAB lists possible device or file completions. ESC at any time exits.

```
grub> ls
(hd0) (hd0,gpt9) (hd0,gpt2) (hd0,gpt1) (fd0)
grub> _
```

GNU GRUB version 2.02~beta2-9ubuntu1

```
*Ubuntu
Advanced options for Ubuntu
Memory test (memtest86+)
Memory test (memtest86+, serial console 115200)
Windows 7 (loader) (on /dev/sda1)
```

Use the ↑ and ↓ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands
before booting or 'c' for a command-line.

A **bootloader** loads the Linux kernel into memory.

Most Linux systems use:

GRUB (Grand Unified Bootloader)

Bootloader Location

Stored in:

- **MBR (Legacy BIOS systems)**
- **EFI partition (UEFI systems)**

Responsibilities

1. Displays **boot menu**
2. Allows user to choose:
 - Linux
 - Recovery mode
 - Another OS (Windows)
3. Loads:
 - **Kernel**
 - **initramfs**

Important GRUB Files

File	Description
<code>/boot/grub/grub.cfg</code>	GRUB configuration
<code>/etc/default/grub</code>	Default settings
<code>/boot/grub2/</code>	GRUB files (RHEL/CentOS)

Example GRUB Menu

Ubuntu
Advanced options for Ubuntu
Memory Test
Windows Boot Manager

3. Kernel Initialization

Kernel Initialization

- Bootloader loads the vmlinuz kernel image file into memory and extracts the contents of the initramfs image file into a temporary, memory-based file system (tmpfs).
- The initial RAM disk (initrd) is an initial root file system that is mounted before the real root file system.
- Initramfs contains the information of kernel modules.
- The job of the initial RAM file system is to preload the block device modules, such as for IDE, SCSI, or RAID, so that the root file system, on which those modules normally reside, can then be accessed and mounted.
- Then the kernel initializes all the hardware for which it can find the drivers in the initramfs, then executes **/sbin/init** from the **initramfs** as **PID 1** in Redhat 7 the initramfs contains a working copy of **systemd** as **/sbin/init**, as well as a udev daemons.
- The initramfs is bound to the kernel and the kernel mounts this initramfs as part of a two-stage boot process.
- The dracut utility creates initramfs whenever a new kernel is installed.
- Use the **lsinitrd** command to view the contents of the image created by Dracut. **/etc/dracut.conf**

```
[ OK ] Started Apply Kernel Variables.
[ OK ] Mounted Kernel Debug File System.
[ OK ] Mounted Huge Pages File System.
[ OK ] Mounted POSIX Message Queue File System.
[ OK ] Started Read and set NIS domainname from /etc/sysconfig/network.
[ OK ] Activated swap /dev/mapper/cl-swap.
[ OK ] Reached target Swap.
[ OK ] Started Remount Root and Kernel File Systems.
      Starting Flush Journal to Persistent Storage...
      Starting Load/Save Random Seed...
      Starting Create Static Device Nodes in /dev...
[ OK ] Started Load/Save Random Seed.
[ OK ] Started Flush Journal to Persistent Storage.
[ OK ] Started Setup Virtual Console.
[ OK ] Started Create Static Device Nodes in /dev.
      Starting udev Kernel Device Manager...
[ OK ] Started udev Kernel Device Manager.
[ OK ] Created slice system-lvm2\x2dpscan.slice.
      Starting LVM event activation on device 8:2...
[ OK ] Started Monitoring of LVM2 mirrors, snapshots etc. using dmeventd or progress polling.
[ OK ] Reached target Local File Systems (Pre).
      Starting File System Check on /dev/disk/by-uuid/0868ca58-6212-404b-91d8-b512c612c58a...
[ OK ] Started LVM event activation on device 8:2.
[ OK ] Started File System Check on /dev/disk/by-uuid/0868ca58-6212-404b-91d8-b512c612c58a.
      Mounting /boot...
[ OK ] Mounted /boot.
[ OK ] Reached target Local File Systems.
      Starting Import network configuration from initramfs...
      Starting Tell Plymouth To Write Out Runtime Data...
      Starting Restore /run/initramfs on shutdown...
[ OK ] Started Restore /run/initramfs on shutdown.
[ OK ] Started Tell Plymouth To Write Out Runtime Data.
[ OK ] Started Import network configuration from initramfs.
      Starting Create Volatile Files and Directories...
[ OK ] Started Create Volatile Files and Directories.
      Starting Security Auditing Service...
```



```

[    0.000000] Linux version 5.11.0-46-generic (buildd@lgw01-amd64-010) (gcc (Ubuntu 9.3.0-17ubuntu1~20.04) 9.3.0, GNU ld (GNU Binutils for Ubuntu) 2.34) #51~20.04.1-Ubuntu SMP Fri Jan 7 06:51:40 UTC 2022 (Ubuntu 5.11.0-46.51~20.04.1-generic 5.11.22)
[    0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-5.11.0-46-generic root=UUID=0b7d1311-b8b7-4dc3-844e-f23666cc878e ro quiet splash
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
[    0.000000]   Hygon HygonGenuine
[    0.000000]   Centaur CentaurHauls
[    0.000000]   zhaoxin   Shanghai
[    0.000000] [Firmware Bug]: TSC doesn't count with P0 frequency!
[    0.000000] x86/fpu: Enabled xstate features 0x7, context size is 832 bytes, using 'standard' format.
[    0.000000] BIOS-provided physical RAM map:
[    0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000009fbff] usable
[    0.000000] BIOS-e820: [mem 0x0000000000009fc00-0x0000000000009ffff] reserved
:

```

The **Kernel** is the **core of the Linux operating system**.

When GRUB loads the kernel, the following happens:

Kernel Responsibilities

1. **Decompress Kernel**
2. **Initialize hardware**
3. **Mount root filesystem**
4. **Load drivers**
5. **Start init process**

Kernel Files

Usually located in:

/boot/vmlinuz

Example:

/boot/vmlinuz-5.14.0

initramfs

Kernel loads a temporary filesystem called:

initramfs

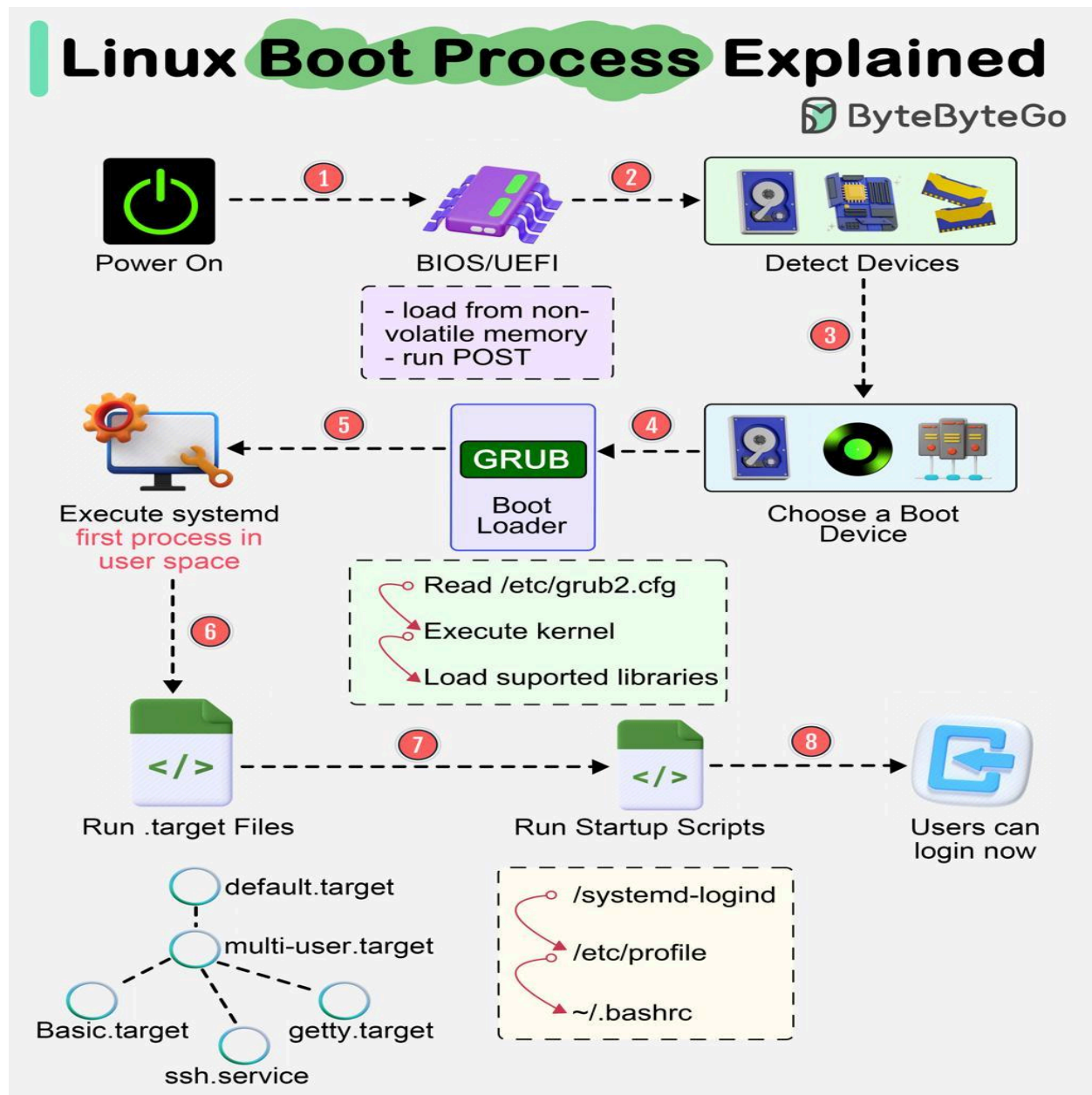
Purpose:

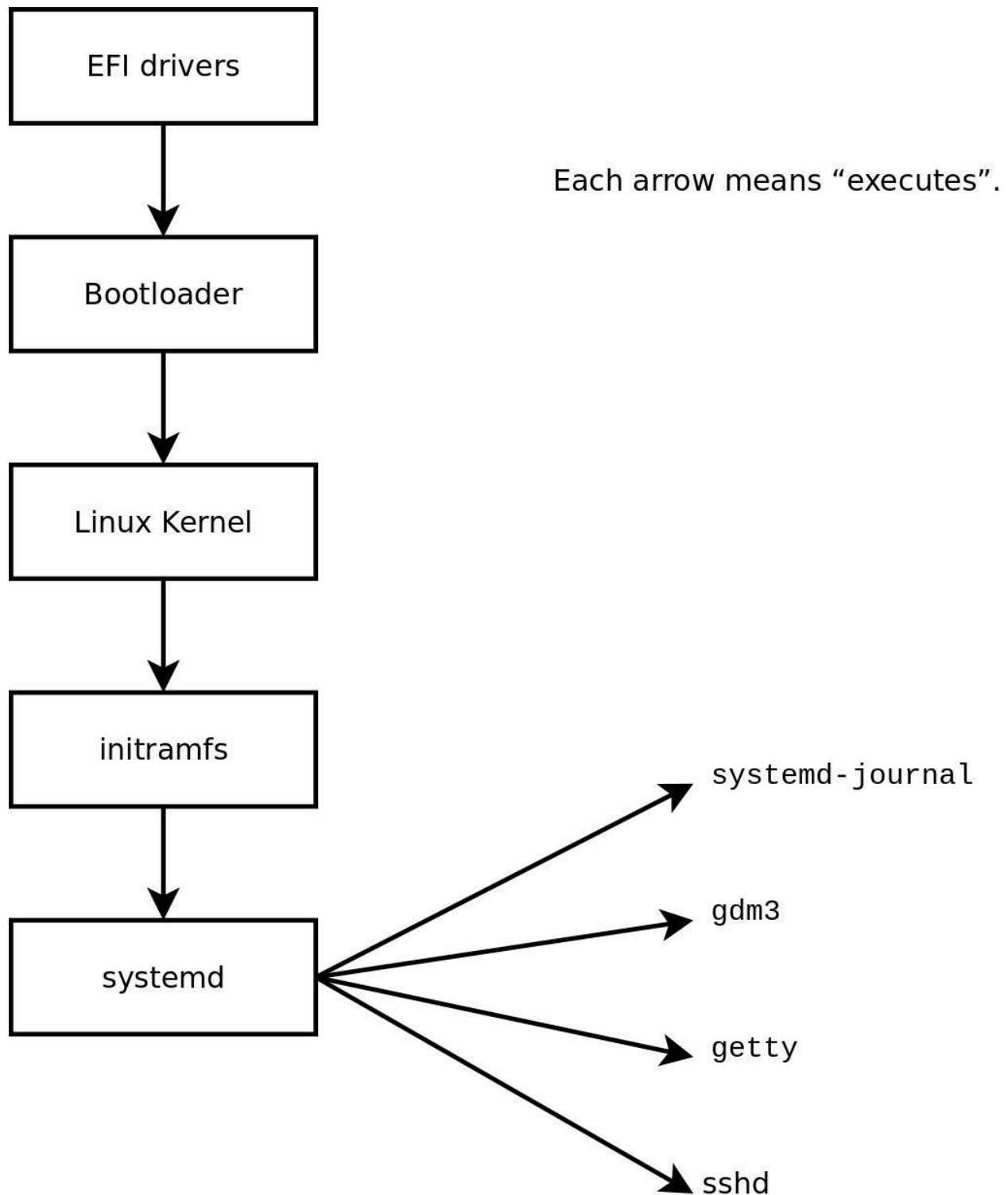
- Contains essential drivers
- Helps mount the real root filesystem

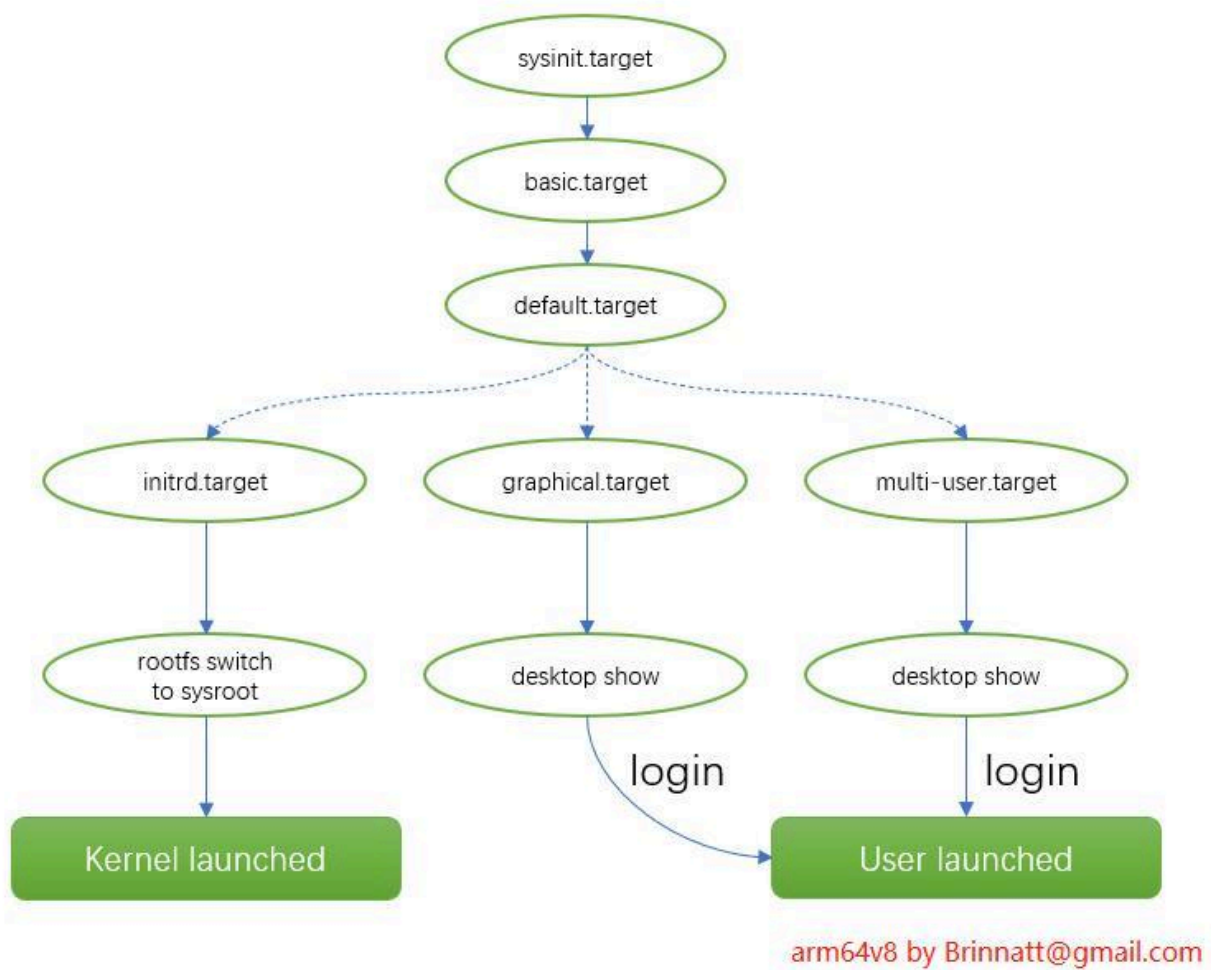
Example:

/boot/initramfs-5.14.0.img

4. init / systemd Process







After the kernel loads, it starts the **first user-space process**.

This process always has:

PID = 1

Traditionally this was **init**, but modern Linux uses **systemd**.

systemd Responsibilities

1. Starts system services
2. Mounts filesystems
3. Starts networking
4. Starts login services

Check PID 1

```
ps -p 1
```

Example output:

```
PID TTY TIME CMD
1 ? 00:00 systemd
```

5. System Initialization (Services Start)

During this stage system services start.

Important Services

Examples:

Service	Purpose
NetworkManager	Network
sshd	SSH access
crond	Scheduled jobs
firewalld	Firewall
httpd	Web server

systemd Targets

Targets replace runlevels.

Target	Description
poweroff.target	Shutdown
rescue.target	Single user
multi-user.target	CLI mode
graphical.target	GUI mode

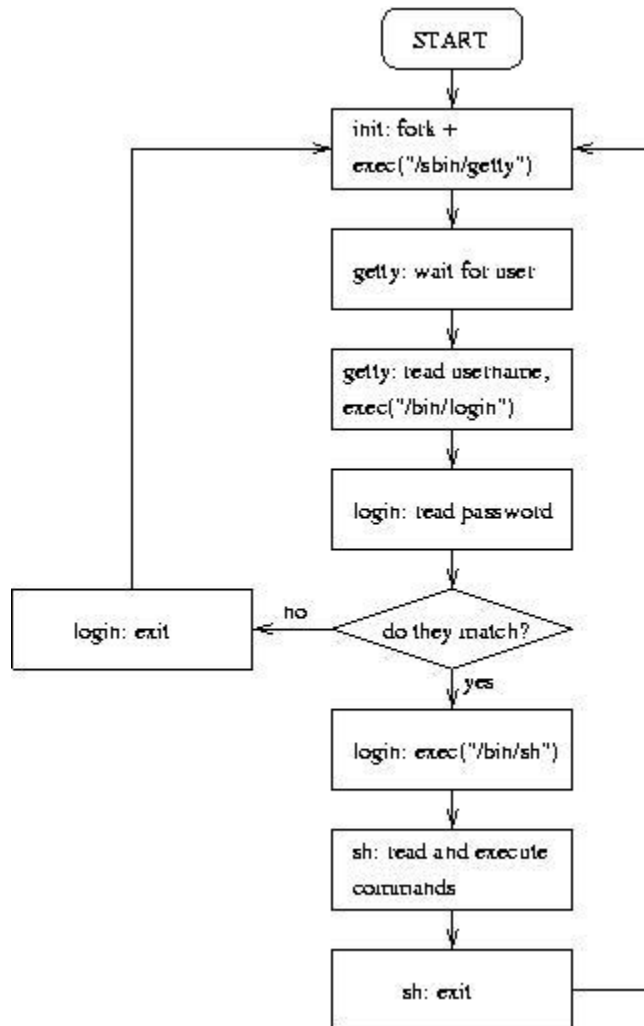
Check Default Target

```
systemctl get-default
```

Example:

graphical.target

6. Login Prompt



```
Ubuntu 18.04.5 LTS ubuntu1804 tty1
ubuntu1804 login: system76
Password:
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 5.4.0-58-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Canonical Livepatch is available for installation.
   - Reduce system reboots and improve kernel security. Activate at:
     https://ubuntu.com/livepatch

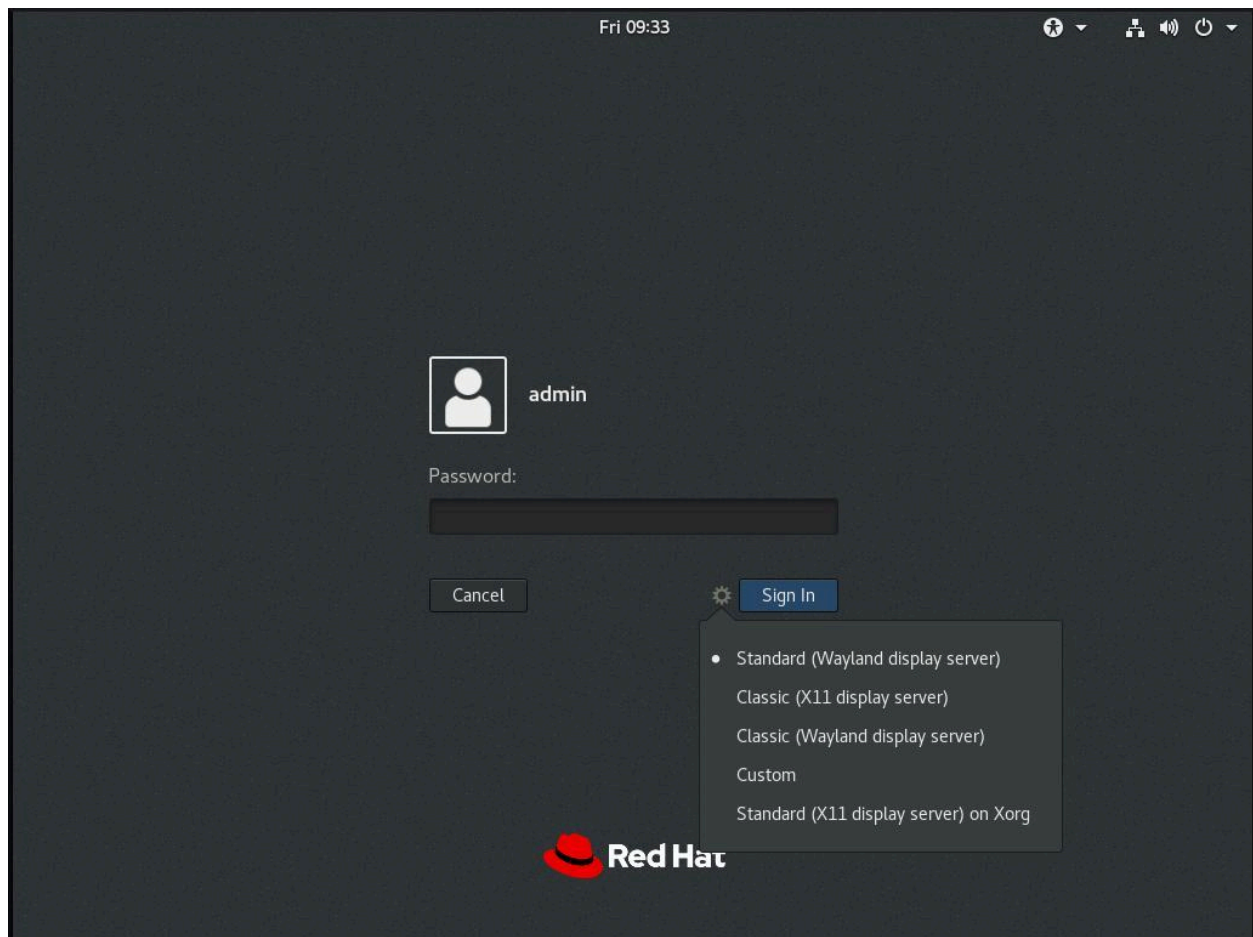
0 packages can be updated.
0 updates are security updates.

Your Hardware Enablement Stack (HWE) is supported until April 2023.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

system76@ubuntu1804:~$ _
```



Finally the system displays the **login prompt**.

User can login through:

CLI Login

login:

password:

GUI Login

Example:

- GNOME
 - KDE
 - LightDM
-

Complete Boot Flow (Quick Overview)

Power ON

↓

BIOS / UEFI

↓

POST (Hardware Check)

↓

Bootloader (GRUB)

↓

Kernel Loaded

↓

initramfs Loaded

↓

systemd (PID 1)

↓

System Services Start

↓

Login Prompt

Important Boot Files

File	Purpose
<code>/boot/vmlinuz</code>	Kernel
<code>/boot/initramfs.img</code>	Temporary root filesystem
<code>/boot/grub/grub.cfg</code>	Bootloader config
<code>/etc/default/grub</code>	GRUB settings
<code>/etc/systemd/</code>	systemd configuration

Linux Boot Process Workflow Explanation

The **Linux boot process** is the sequence of steps that occur when a computer starts and loads the Linux operating system until the system is ready for the user to log in.

The Linux boot process mainly consists of **six stages**:

1. BIOS or UEFI
2. Bootloader
3. Kernel Initialization
4. init or systemd
5. System Services Initialization
6. Login Prompt

Now let us understand each stage one by one.

Stage 1: BIOS or UEFI

When we **power on the computer**, the first program that runs is **BIOS**, which stands for **Basic Input Output System**.

In modern systems, BIOS is replaced by **UEFI**, which stands for **Unified Extensible Firmware Interface**.

The main responsibility of BIOS or UEFI is to perform **POST**, which means **Power-On Self Test**.

During this step, the system checks whether the hardware components are working properly. For example:

- RAM
- CPU
- Keyboard
- Storage devices like HDD or SSD

After checking the hardware, BIOS searches for a **bootable device** based on the configured **boot order**.

Common boot devices include:

- Hard disk
- SSD
- USB drive
- CD or DVD

Once the bootable device is found, BIOS loads the **bootloader** from the disk.

Stage 2: Bootloader

The **bootloader** is responsible for loading the Linux operating system into memory.

The most commonly used bootloader in Linux is called **GRUB**, which stands for **Grand Unified Bootloader**.

GRUB displays a **boot menu** where the user can select which operating system to start.

For example:

- Linux
- Recovery mode
- Windows in dual-boot systems

After the selection is made, GRUB loads two important components:

- The **Linux Kernel**
- The **initramfs** file

Then control is transferred to the kernel.

Stage 3: Kernel Initialization

The **kernel** is the **core component of the Linux operating system**.

Once the kernel is loaded into memory, it starts initializing the system.

The kernel performs several important tasks:

- Detects hardware devices
- Loads device drivers
- Initializes memory management
- Mounts the root filesystem

At this stage, the kernel also loads a temporary filesystem called **initramfs**, which contains essential drivers required to access the real root filesystem.

After completing initialization, the kernel starts the first user-space process.

Stage 4: init or systemd

The first process started by the kernel is called **init**, and it always has **Process ID 1**, also known as **PID 1**.

In older Linux systems, this process was **init**, but in modern Linux distributions it has been replaced by **systemd**.

Systemd is responsible for managing system startup and services.

Its responsibilities include:

- Starting system services
- Mounting filesystems
- Initializing networking
- Managing background processes

Systemd then determines which **target or runlevel** the system should start in.

Stage 5: System Services Initialization

In this stage, system services and background processes are started.

Examples of services include:

- Networking services
- SSH service
- Cron jobs
- Firewall services
- Web servers

Systemd starts these services in the correct order based on dependencies.

Linux systems can run in different **targets**, such as:

- **multi-user.target** for command-line systems
- **graphical.target** for systems with a graphical interface

Once all required services are started, the system moves to the final stage.

Stage 6: Login Prompt

After all services are started successfully, the system displays the **login prompt**.

Users can log in either through:

- **Command Line Interface (CLI)**
or
- **Graphical User Interface (GUI)**

After entering the correct **username and password**, the user gains access to the Linux system and can start working.

Conclusion

So to summarize, the **Linux Boot Process** follows these steps:

1. BIOS or UEFI performs hardware checks.
2. Bootloader loads the Linux kernel.
3. Kernel initializes the system hardware.
4. systemd or init starts the first process.
5. System services are started.
6. Finally, the login prompt appears.

This entire process ensures that the operating system starts correctly and the system is ready for use.
